EXTENSION WORKBOOK

Algebra Extension

Take your algebra to the next level.

Name: _____ Class: _____

My progress

	Skill	Practice done	Skill Check
1	**Factorising Fully** I can take out the highest common factor, e.g. $12x^2+30x = 6x(2x+5)$	☐	/4
2	**Substituting Negatives** I can substitute negative numbers, e.g. if $x = -3$ then $x^2 - 2x = 15$	☐	/4
3	**Index Laws** I can use the index laws and explain them, e.g. $3^4 \times 3^2 = 3^6$	☐	/3
4	**Equations with Brackets** I can solve equations like $3(x + 4) = 27$, and write equations from a story	☐	/3
5	**Inequalities** I can solve an inequality and show it on a number line, e.g. $2x+3 < 11$ gives $x < 4$	☐	/2
6	**Writing Linear Rules** I can write a rule like $y = 2x + 3$ from a table of values or a story	☐	/2
		Skill Check total	/18

1 Factorising Fully

LEARN

Goal: I can take out the highest common factor, e.g. $12x^2 + 30x = 6x(2x + 5)$

Factorising fully means taking out the **highest common factor** (HCF): the **biggest number** and **any letters** that go into every term.

The common mistake. $12x + 36 = 6(2x + 6)$ is **not** fully factorised — look inside the bracket: $2x$ and 6 still have a common factor of 2. Taking out 12 straight away gives $12(x + 3)$.

How to factorise fully

1. **Numbers:** find the HCF. Try the **smaller** number first — does it divide the other one? If not, try its next-biggest factor.
2. **Letters:** if **every** term has the same letter, take it out too (the lowest power).
3. **Divide** each term by the HCF and write the results in the bracket.
4. **Check:** nothing in the bracket shares a factor any more, and expanding gives the question.

Worked examples

$12x + 36$

12 goes into 36, so the HCF is 12

$12x \div 12 = x$, $36 \div 12 = 3$

$= \mathbf{12(x + 3)}$

$18a - 24$

18 doesn't go into 24; 6 does

$18a \div 6 = 3a$, $24 \div 6 = 4$

$= \mathbf{6(3a - 4)}$

$12x^2 + 30x$

Numbers: HCF is 6

Letters: both have x

$= \mathbf{6x(2x + 5)}$

$8ab + 12a$

Numbers: HCF is 4

Letters: both have a

$= \mathbf{4a(2b + 3)}$

$15x^2 - 25x$

Numbers: HCF is 5

Letters: both have x

$= \mathbf{5x(3x - 5)}$

$14xy + 21y$

Numbers: HCF is 7

Letters: both have y

$= \mathbf{7y(2x + 3)}$

Remember: $12x^2 + 30x = 6(2x^2 + 5x)$ is only **partly** factorised: $2x^2$ and $5x$ both still have an x. Always look inside your bracket before you stop.

1 Factorising Fully

PRACTICE

Factorise fully. Show your working in each box.

1 $10x + 30$

$=$ _____

2 $16a - 24$

$=$ _____

3 $18y + 27$

$=$ _____

4 $24x - 36$

$=$ _____

5 $20m + 50$

$=$ _____

6 $42 - 28x$

$=$ _____

7 $x^2 + 9x$

$=$ _____

8 $6x^2 + 9x$

$=$ _____

9 $10a^2 - 15a$

$=$ _____

10 $16x^2 + 24x$

$=$ _____

11 $9ab + 12a$

$=$ _____

12 $20xy - 30y$

$=$ _____

✓ **Check your answers at the back. Fix any you got wrong.**

2 Substituting Negatives

LEARN

Goal: I can substitute negative numbers, e.g. if $x = -3$ then $x^2 - 2x = 15$

When you substitute a **negative number**, always put it in **brackets**. Then use the sign rules and BEDMAS.

Rule	Example
Same signs multiply to a **positive**	$(-3) \times (-4) = 12$
Different signs multiply to a **negative**	$5 \times (-2) = -10$
Squaring a negative gives a **positive**	$(-6)^2 = (-6) \times (-6) = 36$
Subtracting a negative means **add**	$7 - (-2) = 7 + 2 = 9$

Why the brackets matter. If $x = -4$, then x^2 means $(-4)^2 = 16$. Without brackets it is easy to write -4^2 and get -16 by mistake.

Worked examples

$2x + 5$

when $x = -3$

$2 \times (-3) + 5$

$= -6 + 5$

$= -1$

$10 - 2x$

when $x = -3$

$10 - 2 \times (-3)$

$= 10 - (-6) = 10 + 6$

$= 16$

x^2

when $x = -4$

$(-4) \times (-4)$

Same signs: positive

$= 16$

ab

when $a = -3$, $b = -4$

$(-3) \times (-4)$

Same signs: positive

$= 12$

$3a - b$

when $a = -2$, $b = -5$

$3 \times (-2) - (-5)$

$= -6 + 5$

$= -1$

$x^2 - 2x$

when $x = -3$

$(-3)^2 - 2 \times (-3)$

$= 9 + 6$

$= 15$

Remember: Write out every step with brackets. Most mistakes come from doing two sign changes in your head at once.

2 Substituting Negatives

PRACTICE

Find the value of each expression. Show your working in each box.

1 $x + 8$ when $x = -3$

$=$ _____

2 $4x$ when $x = -5$

$=$ _____

3 $7 - x$ when $x = -2$

$=$ _____

4 $3y + 10$ when $y = -4$

$=$ _____

5 a^2 when $a = -6$

$=$ _____

6 ab when $a = -2$, $b = 7$

$=$ _____

7 $2a - b$ when $a = 3$, $b = -4$

$=$ _____

8 $5 - 2x$ when $x = -3$

$=$ _____

9 $x^2 + 3x$ when $x = -2$

$=$ _____

10 $2x^2$ when $x = -3$

$=$ _____

11 $ab - c$ when $a = -2$, $b = -3$, $c = -5$

$=$ _____

12 $3x^2 - 2x$ when $x = -1$

$=$ _____

✓ **Check your answers at the back. Fix any you got wrong.**

3 Index Laws

LEARN

Goal: I can use the index laws and explain them, e.g. $3^4 \times 3^2 = 3^6$

3^4 means $3 \times 3 \times 3 \times 3$. The 3 is the **base** and the 4 is the **index** (power). The index laws only work when the **bases are the same**.

Law	Rule	Example
Multiplying: **add** the indices	$a^m \times a^n = a^{m+n}$	$x^5 \times x^3 = x^8$
Dividing: **subtract** the indices	$a^m \div a^n = a^{m-n}$	$a^7 \div a^2 = a^5$
Power of a power: **multiply** the indices	$(a^m)^n = a^{mn}$	$(x^3)^2 = x^6$

The **base stays the same**. Numbers in front are multiplied or divided as normal: $4x^2 \times 3x^5 = 12x^7$.

Liam says that $3^4 \times 3^2 = 9^6$ because $3 \times 3 = 9$ and $4 + 2 = 6$. Explain why Liam is wrong.

Answer: When you multiply powers with the **same base**, you **keep the base** and **add the indices**: $a^m \times a^n = a^{m+n}$. Liam multiplied the bases, but the base should stay as 3. $3^4 \times 3^2 = (3 \times 3 \times 3 \times 3) \times (3 \times 3)$, which is six 3s multiplied together, so the correct answer is 3^6 $(= 729)$, not 9^6.

Worked examples

$2x^3 \times 5x^4$

Numbers: $2 \times 5 = 10$

Indices: $3 + 4 = 7$

$= \mathbf{10x^7}$

$15a^6 \div 3a^2$

Numbers: $15 \div 3 = 5$

Indices: $6 - 2 = 4$

$= \mathbf{5a^4}$

$(2x^3)^2$

Everything in the bracket is squared

$2^2 = 4$, $3 \times 2 = 6$

$= \mathbf{4x^6}$

Remember: Watch out for these common mistakes:

- $2x^3 \times 5x^4$ is $10x^7$, not $7x^7$ — the numbers in front are **multiplied**, only the indices are added.
- $(2x^3)^2$ is $4x^6$, not $2x^6$ — the 2 inside the bracket is squared too.
- $x^6 \div x^2$ is x^4, not x^3 — dividing powers means **subtract** the indices, not divide them.

3 Index Laws

PRACTICE

Simplify, leaving your answer in index form. Show your working in each box.

1 $5^3 \times 5^4$

$=$ _____

2 $x^6 \times x^2$

$=$ _____

3 $7^9 \div 7^3$

$=$ _____

4 $a^{10} \div a^4$

$=$ _____

5 $(x^4)^3$

$=$ _____

6 $3x^2 \times 4x^5$

$=$ _____

7 $20a^8 \div 5a^3$

$=$ _____

8 $(3x^2)^2$

$=$ _____

9 Mia says $2^3 \times 2^4 = 4^7$ because $2 \times 2 = 4$ and $3 + 4 = 7$. Explain why Mia is wrong. State the index law and give the correct answer.

_____

_____

_____

_____

10 Tane says $5^8 \div 5^2 = 1^4$ because $5 \div 5 = 1$ and $8 \div 2 = 4$. Explain why Tane is wrong. State the index law and give the correct answer.

_____

_____

_____

_____

✓ **Check your answers at the back. Fix any you got wrong.**

4 Equations with Brackets

LEARN

Goal: I can solve equations like $3(x+4)=27$, and write equations from a story

Method 1 — expand first. Expand the bracket, then solve the two-step equation.
Method 2 — divide first. If the number outside divides the other side exactly, divide first.

$3(x+4)=27$

Expand: $3x+12=27$

$3x=15$

$= x = 5$

$3(x+4)=27$

Divide by 3: $x+4=9$

Take 4 from both sides

$= x = 5$

$2(3x-1)=19$

Expand: $6x-2=19$

$6x=21$

$= x = 3.5$

Check by substituting your answer back in: $3(5+4)=3\times 9=27$ ✓

Writing equations from a story

Turn words into an equation

1. Choose a **letter** for the unknown (say what it stands for).
2. Follow the story **in order**. Use **brackets** when something happens to a whole amount: "add 5, **then** multiply by 4" is $4(n+5)$.
3. Make it **equal** to the total, then **solve**.

Sione's number

Thinks of a number n, adds 5, then multiplies by 4. The answer is 36.

$4(n+5)=36$

$= n = 4$

Rectangle

Width 5 cm, length $(x+3)$ cm, area 45 cm².

$5(x+3)=45$

$= x = 6$

Pizza

3 friends each buy a pizza ($\$p$) and a \$4 drink. Total \$45.

$3(p+4)=45$

$= p = 11$

4 Equations with Brackets

PRACTICE

Solve each equation. For the stories, write an equation and then solve it. Show your working in each box.

1 $2(x + 3) = 14$

$x =$ _____

2 $5(x - 2) = 35$

$x =$ _____

3 $4(2x + 1) = 36$

$x =$ _____

4 $3(x + 5) = 6$

$x =$ _____

5 $6(x - 4) = -12$

$x =$ _____

6 $2(3x - 1) = 19$

$x =$ _____

7 Aroha thinks of a number, subtracts 3, then multiplies the result by 5. The answer is 40. Write an equation and solve it to find her number.

Equation: _____ Answer: _____

8 A rectangle is 4 cm wide and $(x + 6)$ cm long. Its area is 52 cm². Write an equation and solve it to find x.

Equation: _____ Answer: _____

9 Five students each buy a movie ticket costing $\$t$ and a \$3 popcorn. They spend \$85 altogether. Write an equation and solve it to find the cost of a ticket.

Equation: _____ Answer: _____

✓ **Check your answers at the back. Fix any you got wrong.**

5 Inequalities

LEARN

Goal: I can solve an inequality and show it on a number line, e.g. $2x + 3 < 11$ gives $x < 4$

Symbol	Meaning	Circle on the number line
$<$	less than	open ○
$>$	greater than	open ○
$\leq$	less than **or equal to**	filled ●
$\geq$	greater than **or equal to**	filled ●

The **open** circle means the number itself is **not** included; **filled** means it is. The arrow points the way the numbers go: **left** for less than, **right** for greater than.

How to solve an inequality
Solve it **exactly like an equation**, with one extra rule: if you **multiply or divide by a negative number**, **flip the inequality sign**.

Worked examples

1. $2x + 3 < 11 \quad \Rightarrow \quad 2x < 8 \quad \Rightarrow \quad x < 4$ (open circle, arrow left)



2. $3(x - 1) \geq 6 \quad \Rightarrow \quad x - 1 \geq 2 \quad \Rightarrow \quad x \geq 3$ (filled circle, arrow right)



3. $5 - 2x \leq 11 \quad \Rightarrow \quad -2x \leq 6 \quad \Rightarrow \quad x \geq -3$ (divided by -2, so **flip**!)



Remember: Check with a test number. For $x \geq -3$, try $x = 0$: $5 - 2 \times 0 = 5$, and $5 \leq 11$ ✓. So numbers bigger than -3 work, and the arrow points right.

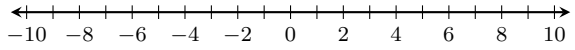
5 Inequalities

PRACTICE

Solve each inequality, then show the answer on the number line. Show your working in each box.

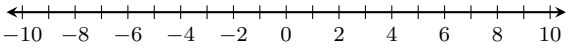
1 $x + 4 > 7$

Answer: _____



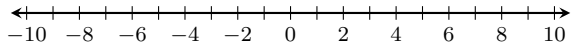
2 $3x \leq 12$

Answer: _____



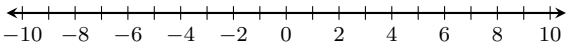
3 $2x - 5 < 7$

Answer: _____



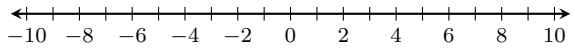
4 $4x + 3 \geq -9$

Answer: _____



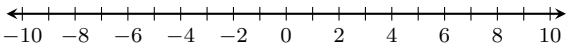
5 $\frac{x}{2} + 1 > 3$

Answer: _____



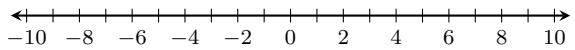
6 $3(x - 2) \leq 9$

Answer: _____



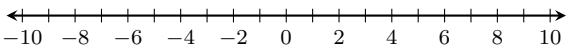
7 $5 - x < 2$

Answer: _____



8 $-2x + 1 \geq 7$

Answer: _____



✓ **Check your answers at the back. Fix any you got wrong.**

6 Writing Linear Rules

LEARN

Goal: I can write a rule like $y = 2x + 3$ from a table of values or a story

A linear rule looks like $y = mx + c$.

- m is how much y **changes** each time x goes up by 1 (the **difference**).
- c is the value of y when $x = 0$ (the **starting value**).

From a table

1. Check x goes up by 1 each time. Find the **difference** between the y values: that's m.
2. Find y when $x = 0$: that's c. If the table doesn't start at 0, **work backwards** one step at a time.
3. Write $y = mx + c$ and **check** it with another column.

x	0	1	2	3
y	3	5	7	9

Goes up by 2: $m = 2$

At $x = 0$, $y = 3$: $c = 3$

$= y = 2x + 3$

x	1	2	3	4
y	7	10	13	16

Goes up by 3: $m = 3$

Back one step: $7 - 3 = 4$

$= y = 3x + 4$

x	0	1	2	3
y	10	8	6	4

Goes **down** by 2: $m = -2$

At $x = 0$, $y = 10$

$= y = -2x + 10$

From a story

The **fixed amount** (a starting fee or starting value) is c. The amount **per** hour, week, km... is m.

Gym

\$60 joining fee plus \$15 per week. Cost C after w weeks.

$= C = 15w + 60$

Candle

30 cm tall, burns down 2 cm every hour. Height h after t hours.

$= h = 30 - 2t$

Phone

\$20 a month plus \$5 per GB of data. Cost C for g GB.

$= C = 5g + 20$

6 Writing Linear Rules

PRACTICE

Write the rule for each table. For the stories, write an equation. Show your working in each box.

1

x	0	1	2	3
y	1	4	7	10

$y =$ _____

2

x	0	1	2	3
y	5	7	9	11

$y =$ _____

3

x	1	2	3	4
y	6	10	14	18

$y =$ _____

4

x	1	2	3	4
y	2	7	12	17

$y =$ _____

5

x	0	1	2	3
y	12	9	6	3

$y =$ _____

6

x	2	3	4	5
y	1	3	5	7

$y =$ _____

7 An electrician charges a \$70 call-out fee plus \$45 per hour. Write an equation for the total cost C of a job that takes h hours.

_____

8 A taxi charges \$3.50 to start, plus \$2 per kilometre. Write an equation for the fare F for a trip of d kilometres.

_____

9 A water tank holds 200 litres and leaks 5 litres every hour. Write an equation for the amount of water W left after t hours.

_____

✓ Check your answers at the back. Fix any you got wrong.

✓ Skill Checks

TEACHER MARKS

Name: _____

Do these **on your own — no notes**. Show your working, and write your final answer on the line.

1 Factorising Fully

Factorise fully. **Mark: ____ / 4**

1 $15x + 45$

$=$ _____

2 $24a - 36$

$=$ _____

3 $8x^2 + 20x$

$=$ _____

4 $18xy - 12x$

$=$ _____

2 Substituting Negatives

Find the value. **Mark: ____ / 4**

1 $5x + 3$ when $x = -2$

$=$ _____

2 $a - b$ when $a = -4,\ b = -6$

$=$ _____

3 $x^2 - 4x$ when $x = -3$

$=$ _____

4 $2ab$ when $a = -5,\ b = 3$

$=$ _____

3 Index Laws

Simplify, and explain. **Mark: ____ / 3**

1 $2^5 \times 2^3$

$=$ _____

2 $12x^7 \div 3x^2$

$=$ _____

3 Ana says $4^3 \times 4^5 = 16^8$ because $4 \times 4 = 16$ and $3 + 5 = 8$. Explain why Ana is wrong. State the index law and give the correct answer.

_____

_____

_____

Skill Checks

CONTINUED

Name: _____

Total: ____ / 18

Do these **on your own — no notes**. Show your working, and write your final answer on the line.

4 Equations with Brackets

Solve.

Mark: ____ / 3

1

$3(x - 4) = 18$

$x =$ _____

2

$4(2x + 3) = 4$

$x =$ _____

3

Tama thinks of a number, adds 7, then multiplies by 3. The answer is 39. Write an equation and solve it.

Equation: _____

Answer: _____

5 Inequalities

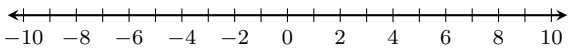
Solve, then show on the number line.

Mark: ____ / 2

1

$3x - 4 > 8$

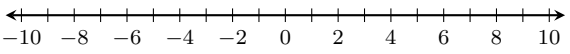
Answer: _____



2

$6 - 2x \leq 10$

Answer: _____



6 Writing Linear Rules

Write the rule.

Mark: ____ / 2

1

x	1	2	3	4
y	1	4	7	10

$y =$ _____

2

A school trip costs $200 for the bus plus $12 per student. Write an equation for the total cost C when n students go.

_____

Answers to the Practice Questions

Terms can be in a different order and still be right — e.g. $2b + 3a$ is the same as $3a + 2b$.

1. Factorising Fully

1. $10(x + 3)$
2. $8(2a - 3)$
3. $9(2y + 3)$
4. $12(2x - 3)$
5. $10(2m + 5)$
6. $14(3 - 2x)$
7. $x(x + 9)$
8. $3x(2x + 3)$
9. $5a(2a - 3)$
10. $8x(2x + 3)$
11. $3a(3b + 4)$
12. $10y(2x - 3)$

2. Substituting Negatives

1. 5
2. -20
3. 9
4. -2
5. 36
6. -14
7. 10
8. 11
9. -2
10. 18
11. 11
12. 5

3. Index Laws

1. 5^7
2. x^8
3. 7^6
4. a^6
5. x^{12}
6. $12x^7$
7. $4a^5$
8. $9x^4$
9. When multiplying powers with the same base, keep the base and add the indices: $a^m \times a^n = a^{m+n}$. The base stays 2, so $2^3 \times 2^4 = 2^7$ $(= 128)$, not 4^7.
10. When dividing powers with the same base, keep the base and subtract the indices: $a^m \div a^n = a^{m-n}$. The base stays 5, so $5^8 \div 5^2 = 5^6$, not 1^4.

4. Equations with Brackets

1. $x = 4$
2. $x = 9$
3. $x = 4$
4. $x = -3$
5. $x = 2$
6. $x = 3.5$
7. $5(n - 3) = 40$, so $n = 11$
8. $4(x + 6) = 52$, so $x = 7$
9. $5(t + 3) = 85$, so $t = 14$

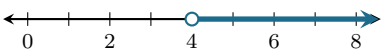
5. Inequalities

1. $x > 3$ 

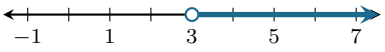
2. $x \leq 4$ 

3. $x < 6$ 

4. $x \geq -3$ 

5. $x > 4$ 

6. $x \leq 5$ 

7. $x > 3$ 

8. $x \leq -3$ 

6. Writing Linear Rules

1. $y = 3x + 1$
2. $y = 2x + 5$
3. $y = 4x + 2$
4. $y = 5x - 3$
5. $y = -3x + 12$
6. $y = 2x - 3$
7. $C = 45h + 70$
8. $F = 2d + 3.5$
9. $W = 200 - 5t$